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Linolium: A Containerized Workflow for Automated Pathogen Lineage Designation and Interactive Curation

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Summary

In public health, pathogen lineage designation is the process of identifying distinct groups within a species which is essential for tracking the transmission of infections and informing treatment strategies(McBroome et al., 2024). Linolium is an end-to-end containerized workflow for assigning and exploring lineage designations for pathogens using genomic data. It orchestrates lineage inference, metadata integration, and browser-based review within a single reproducible environment. Linolium enables users to assign, inspect, and refine lineage systems using genomic data represented in Mutation Annotated Tree (MAT) format(Turakhia et al., 2021), supporting both established nomenclatures and the development of new lineage systems.

Statement of need

Recent surges in pathogen sequence availability have challenged existing methods of lineage detection and curation. Pathogen surveillance methods developed during the SARS-CoV-2 pandemic, including wastewater-based tracking (Gangwar et al., 2025; Karthikeyan et al., 2022), have catalyzed a transition toward high-volume sequencing. Yet, this abundance of data has exposed a bottleneck in our ability to maintain up to date lineages and identify emerging threats, necessitating a more standardized and scalable approach to pathogen genomics. Existing tools address lineage identification and phylogenetic visualization separately, requiring users to manually connect outputs across workflows.

While our research group previously developed individual tools for these tasks (McBroome et al., 2021, 2024; Sanderson, 2022; Turakhia et al., 2021; *Using TaxoniumTools - Taxonium Documentation*, n.d.), users must still navigate fragmented processes to achieve a full analysis. Linolium addresses this gap by automating the multi-tool requirements of lineage designation, as well as contributing specialized visualization and data handling into a streamlined environment. Designed for public health researchers, this tool reduces the manual coordination typically required for large pathogen datasets by integrating inference and curation steps. Linolium combines existing lineage designation and visualization tools and introduces metadata-aware lineage scoring within a unified workflow. The tool lowers technical barriers by encapsulating dependencies and managing intermediate data automatically.

State of the field

Pathogen lineage designation is a decentralized process across many pathogen-specific nomenclatures (Hill et al., 2024; Kuhn et al., 2014; Lancefield, 1933; O'Toole et al., 2021,

2022; Ramaekers et al., 2020; Rambaut et al., 2020; Schneider et al., 2019; Seabra et al., 2022) and a manual suggestion and curation process which is time-consuming and biased by human opinion(Focosi & Maggi, 2023; McBroome et al., 2024). In response to pandemic level sequencing, new tools such as AutoLin and Taxonium were developed which assist with elements of the lineage identification process such as candidate lineage identification and phylogenetic visualization. Despite the development of these tools, there was no tool which supported the identification of new lineages from beginning to end. Linolium orchestrates AutoLin(McBroome et al., 2024), TaxoniumTools(*Using TaxoniumTools - Taxonium Documentation*, n.d.), and Taxonium(Sanderson, 2022), managing dependencies and intermediate data while extending AutoLin with metadata-weighted lineage scoring and integrating interactive curation. Linolium consolidates existing lineage designation and visualization tools into a unified workflow for automated and interactive lineage curation.

AutoLin is a pathogen-agnostic software tool which assists in automated, efficient, and phylogenetic-based lineage designation for faster identification of differentiation within species(McBroome et al., 2024). While AutoLin is highly versatile and scalable across various pathogens, it lacks a function to weight phenotypes defined in metadata which would provide more biologically informed lineage suggestions. Linolium adds additional parameters to AutoLin that allow the user to improve the biological relevance of the proposed lineages (see ./autolin/README.md).

Taxonium is a web-based, interactive visualization tool for large (>10M samples) phylogenetic trees and associated metadata(Sanderson, 2022). The original Taxonium tool is capable of displaying lineage information for nodes within a phylogeny but lacks the ability to identify AutoLin suggested lineages and curate this information interactively. The Linolium implementation of Taxonium has a new user interface that specifically identifies AutoLin outputs and allows for immediate review of the proposed new lineages.

Conversion of data across this analysis requires TaxoniumTools, a useful but technical tool that translates phylogenetic tree files into Taxonium readable files. Because TaxoniumTools has many features and specific use cases that require advanced knowledge and significantly increase technical overhead, Linolium replaces user interaction with this tool, handling all necessary inputs and outputs to convert informational results into interpretable visualizations with no additional commands.

Software Design

Linolium is a containerized workflow that connects analysis, conversion, and visualization tools. While AutoLin and Taxonium are effective independently, their lack of integration necessitates significant manual data transformation and multiple command-line steps. We integrated these tools within a Docker container to ensure reproducibility and simplify dependency management across heterogeneous computing environments common in public health genomics.

Rather than developing competing software, we chose to build upon existing open-source tools to improve their efficiency and biological relevance. Linolium uses data formats and tools that are well documented and commonly used to build on existing frameworks and provide reliable results. All of its analyses and visualizations are built around the UShER Mutation Annotated Tree (MAT) format(Turakhia et al., 2021), which has become widely used in public health applications, tracking infectious disease evolution(Karim et al., 2025; Martínez-Martínez et al., 2023; Mboowa et al., 2026; McBroome et al., 2022; Turakhia et al., 2022). We extended the core logic of AutoLin by introducing a metadata-weighting mechanism that incorporates user-defined contextual variables into lineage scoring and prioritization. This allows the software to accept user-defined metadata, such as geographic or clinical importance, ensuring that the resulting lineage assignments are tailored to the specific biological context of the research.

The Linolium lineage curation browser interface is built with the Taxonium library. We extended Taxonium with lineage-aware visualization components that directly ingest AutoLin proposals

(Figure 1), enable node-level lineage reassignment (Figure 2 A and B), and export updated lineage annotations in a format compatible with downstream phylogenetic analysis (Figure 1). This creates a unified environment in which automated lineage inference and expert-driven revision operate on the same underlying tree representation. This design supports iterative cycles of automated lineage proposal followed by expert curation within the same analytical environment. By centralizing these tools into a unified, containerized application, Linolium consolidates previously separate command-line workflows into a single reproducible system for lineage designation and curation.

Figure 1: Visualization of M. tuberculosis Lineage 4.8 phylogeny with pre-existing and AutoLin assigned lineages within Linolium browser interface. A) The mutation annotated trees and edited lineage information can be exported directly from the browser in JSONL format or edited lineage information can be exported in tsv format. B) All lineages within the MAT can be viewed and searched and descending lineages can be collapsed within viewer. C) Full interactive tree displayed, with ability to zoom on specific clades and nodes.

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Figure 2: A) Linolium browser in lineage reassignment node for AutoLin designated lineage 4.8.62. In this mode any node can be selected as the new parent node for the lineage. B) AutoLin designated Lineage auto.lineage4.8.62 after parent node reassignment. Node reassignment will identify all descendent leaves and assign their correct lineage. In addition to parent node reassignment, a lineage can also be merged into the parent lineage, essentially rejecting a suggested lineage.

Because the ultimate goal of the analysis is interactive exploration, we chose to host all user interaction within a locally-hosted browser-based interface within the Docker container. This design removes the requirement for command-line expertise, allowing users to upload data and manage parameters through a graphical environment. This integrated approach transforms a series of isolated technical tasks into a complete, accessible pathway for pathogen surveillance.

Research Impact Statement

To ensure immediate utility for public health genomic surveillance, the Linolium pipeline is built as a high-performance, reproducible environment that is currently ready for external adoption. A suite of Python command line tools powers the backend, while a locally hosted web interface manages the workflow, enabling users to execute and monitor analysis pipelines sequentially. The web interface, lineage designation tools, and interactive visualization are fully containerized using Docker, allowing for deployment on many computing environments without the need for manual dependency management. Community-readiness is further signaled by the public availability of the source code at https://github.com/corbett-lab/lineage-curation/tree/main, which includes highly documented README and example datasets to facilitate rapid onboarding for new users. By leveraging the UShER framework, Linolium is natively compatible with existing global-scale Mutation Annotated Trees (MATs), making it an immediately viable solution for researchers monitoring Tuberculosis, Monkey Pox, Dengue fever and many many other pathogens.

AI Usage Disclosure

We used generative AI tools, primarily for elements of the web browser interface and developing the Docker container. AI tools were used for editing grammar and wording within the manuscript but not for generating written content. All AI-generated content was carefully reviewed and verified before inclusion.

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